

Project Smart Trigger for ECE/CS 3992

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Abstract—This project proposes a smart trigger for firearms that uses RFID technology to authenticate the user and ensure that only authorized users can fire the gun. This would help to prevent accidental shootings and unauthorized use of firearms, ultimately improving public safety. This paper will discuss solutions already in the market, the proposed solution, and the ethical considerations that need to be taken into account when designing and implementing such a system. It will also explore the project and outline the tasks that need to be completed in order to successfully implement the smart trigger and demonstrate its functionality at Demo Day. The risks associated with the project will also be assessed and strategies for mitigating those risks will be discussed.

Index Terms—IEEE, IEEEtran, journal, L^AT_EX, paper, template.

I. PROBLEM STATEMENT

A huge problem with firearms is that they are dangerous and can cause serious harm or death if not handled properly. This is especially true when it comes to children getting ahold of firearms or unauthorized users who may not know how to handle a firearm safely. Police officers can lose their firearms in the line of duty and if they are not recovered they can be used in crime. The proposed system, a smart trigger, would help to address this problem by preventing the gun from firing unless it is used by the owner. It would use RFID technology to authenticate the user and ensure that only authorized users can fire the gun. This would help to prevent accidental shootings and unauthorized use of firearms, ultimately improving public safety.

II. SOLUTION SPACE

Efforts to improve firearm safety through access control have evolved through several distinct technological approaches. At the most basic level, traditional solutions rely on purely mechanical safeguards such as trigger locks or integrated key-based locking mechanisms. These systems are inexpensive and highly reliable because they do not depend on electronics or power, but they fundamentally lack user-specific authentication.

Anyone with access to the key or combination can operate the firearm, which causes concern toward illegitimate use. In time-sensitive or high-stress situations, the need to manually unlock a device with a key can render the safety ineffective in practice, leading many police and military personnel to avoid using such systems altogether. This creates a gap between theoretical safety and real-world adoption, motivating the development of systems that can enforce access control without requiring manual, non-instantaneous intervention.

Electronic authentication methods represent the next step in this progression. Systems based on PIN entry, external key fobs, or smartphone connectivity attempt to introduce user-specific control while maintaining relatively simple hardware implementations. While these approaches improve upon mechanical systems by enabling programmable access, they still require explicit interaction from the user, such as entering a code or activating a device. This introduces latency and cognitive load that can be unacceptable in defensive scenarios. Additionally, wireless-based systems may be vulnerable to interception or spoofing, depending on their implementation. As a result, these approaches highlight the need for authentication mechanisms that are both secure and implicit, allowing the system to verify the user seamlessly without requiring conscious action.

Biometric authentication has emerged as the most prominent alternative in this space, particularly in commercial “smart gun” designs. By leveraging physiological traits such as fingerprints, palm vein patterns, or even grip characteristics, biometric systems aim to bind firearm operation directly to the identity of the user. This approach is compelling because it eliminates the need for external tokens and ensures that only authorized individuals can operate the weapon. However, when examined in the context of real-world firearm use, biometrics introduce a number of critical challenges. Sensors such as fingerprint scanners are highly sensitive to environmental conditions. Metrics like dirt, sweat, gloves, or moisture can significantly degrade performance. Firearms are often used in unpredictable and harsh environments, where such conditions are common, leading to an increased

likelihood of authentication failure.

Latency is another major concern. Biometric systems require time to capture, process, and match data against stored templates. Even small delays can be unacceptable in scenarios where immediate response is necessary. Equally important is the issue of reliability in terms of false negatives and false positives. A false negative, where an authorized user is denied access, represents a potentially dangerous failure in a defensive situation. Conversely, a false positive undermines the entire purpose of the system. These risks are compounded by the long-term degradation of sensors and the variability of human biological signals. In addition to these technical concerns, biometric systems tend to be significantly more expensive and power-intensive than alternative solutions. Commercial implementations often cost on the order of \$1000 to \$1500, reflecting the complexity of most biometric devices. This cost and power burden conflicts directly with the constraints of a small, battery-powered module designed to fit within the firing control group of an AR-15.

RFID-based authentication offers a different point in the design space by prioritizing simplicity, robustness, and low power consumption. In a passive near-field RFID system, authentication occurs when a tag enters the electromagnetic field generated by a reader, allowing the tag to transmit its unique identifier without requiring its own power source. This enables extremely low-latency operation and reduces the system's overall energy requirements. RFID systems are also less sensitive to environmental contaminants compared to optical or capacitive biometric sensors, making them more reliable in varied conditions. The primary tradeoff is that RFID relies on possession rather than inherent identity. A user must carry an external token, such as a ring or card, which introduces the possibility of loss or unauthorized use if the token is compromised.

More complex hybrid systems attempt to combine multiple authentication methods, such as pairing RFID with biometrics or behavioral sensing, in order to improve security. While these approaches can reduce certain vulnerabilities, they significantly increase system complexity, cost, and power consumption. For a compact, retrofittable module operating under strict physical and electrical constraints, this hybrid complexity is impractical. As a result, the solution space reflects a series of tradeoffs between security, reliability, usability, and feasibility. Biometric systems offer strong identity assurance but struggle with robustness and cost, while RFID systems sacrifice some degree of identity binding in exchange for speed, reliability, and practicality. This balance makes RFID a compelling candidate for a system

intended to be compact, cost-effective, and deployable in real-world conditions.

III. RELATED WORK

A more in-depth, formal exploration of the research and product development related to the problem space.

IV. SOLUTION / CONTRIBUTION

1-2 paragraphs summarizing your approach, what's exciting about it, why it will succeed

V. SAFETY, GLOBAL, SOCIETAL, ETHICAL CONSIDERATIONS

What are the public health, safety and welfare, global, cultural, societal, environmental, and economic factors related to the problem and solution space? What power does the proposed system provide, and to whom? What are the relevant professional ethical obligations to consider in the design of the system?

A. Factors Related to the Problem and Solution

1) *Public Health, Safety, and Welfare:* The main issue the proposed system addresses is the potential for negligent discharge of a firearm. This is because of the nature of guns in that they are designed to inflict harm on a target. The smart trigger would help to protect the public safety in a couple of ways. First the system would be designed to prevent the gun from firing unless it is used by the owner. This would help to prevent accidental shootings by children or others that are not licensed to shoot a firearm. Secondly, if the gun was lost unless the module was removed the gun wouldn't be able to fire. This would help to prevent the gun from being used in crime or from being used in a negligent way. Lastly, the system is module meaning that if at any point it has a malfunction the module can be removed and the gun can be used as normal. This would help to prevent situations where a gun is rendered unusable due to malfunctions of the system. In the past since the mid 1990s, there have been numerous starts to create something similar to the smart trigger. These safety devices are known as Firearm Safety Devices (FSDs) and are most of the time mechanical as they are reliable and don't require a power source as shown in fig ??.

They also have some characteristics and features that are mentioned in [?]

- the option of integrating and combining mechanical devices with various authorization processes (token, password, biometric);
- hierarchical authorization;

- traceability and logging functions;
- networking and remote availability.



Fig. 1. Trigger and Cable Lock

2) *Global, and Cultural*: Because our project is focused on firearms it is important to consider the global and cultural factors that may affect the implementation and use of the smart trigger. For example, different countries have different laws and regulations regarding firearms, and the cultural attitudes towards guns can vary significantly. It is important to ensure that the system is designed in a way that is compliant with these regulations and respectful of these cultural differences. For the scope of the project we will be focusing on the United States, but it is also important to consider its implications in other countries. The United States has a deep history with firearms and the Second Amendment to the Constitution guarantees the right of citizens to keep and bear arms. This doesn't mean that no regulations have been put in place regarding firearms. Some laws that may affect the system include.

- Maryland (SB 334/HB 1339) This law requires that all firearms be equipped with a safety device before being sold or transferred. In order for our system to be approved in Maryland it would need to be approved by the Handgun Roster Board, which is responsible for evaluating and approving firearms and firearm safety devices for sale in the state.
- 1) New Jersey's "Display Mandate": This law requires that all firearms be displayed in a visible manner when being sold or transferred. This would require the entire state to adopt the smart trigger as the standard to use in New Jersey.
 - 2) Because our system uses RFID technology its legally and intentional radiator. This means that if we ever wanted to sell the system in the United States we would need to get approval from the Federal Communications Commission (FCC)

to ensure that it complies with regulations regarding radio frequency emissions.

3) *Societal and Economic*: Part of the reason why the RFID authentication would be beneficial is that it would be a more affordable and reliable solution compared to other biometric authentication methods. This would make it more accessible to a wider range of gun owners, which could help to increase the adoption of the system and ultimately improve public safety. At gun ranges unless the user was wearing the RFID tag they wouldn't be able to use the gun. This would make shooting ranges safer for everyone because it would prevent unauthorized users from using the guns. This would also help to prevent accidents and injuries at shooting ranges, which could have a positive impact on public health and safety. Another way in which the system would increase safety is if a gun was lost or stolen from the police department. If the gun was equipped with the smart trigger it would be rendered useless to any thief.

For each of the

Microcontroller	\$15 or free
RFID Scanner	9prototypeboard,3-5 per chip
AR-15 Lower	Free (thanks Stephen)
3D Printed Parts	\$20 maximum
Custom PCB	\$50 in prototyping, ~25 per order
Various electronic parts for the PCBs	\$20-50
Linear Actuator	\$20
RFID Tags	10forprototypes,10 for the ring
RFID Programmer	\$30
Battery	10forbattery,5 for charger parts
Total	\$190-235

4) *Environmental*: The use of RFID technology in our system also has some environmental considerations. The production and disposal of the RFID tags and other electronic components can have an impact on the environment. However, we will strive to use sustainable materials and practices wherever possible to minimize this impact.

B. Power

Understanding how the system could be used and how power is shifting is important to ensure that the system is designed in a way that is ethical and responsible. The smart trigger would provide power to the gun owner by allowing them to have more control over who can use their firearm. This would help to prevent unauthorized use of the gun, which could improve public safety and reduce the risk of accidents and injuries. This would also give power to law enforcement by making it more difficult for stolen guns to be used in crimes and against police officers in shoot outs. The system would also provide power to the general public by helping to prevent

accidents and injuries caused by negligent discharge of firearms. If the system were to have any bugs or issues it could potentially be exploited by moving power to the criminals.

VI. PROPOSED WORK

A project includes some things and excludes others you know you won't work on. Define more explicitly what is included and not included here. What will be demonstrated at project completion at Demo Day?

For our project, we will include parts of the gun that are necessary to demo the smart trigger functionality.

Items Included:

- Lower receiver: This is the part of the gun that contains the trigger mechanism and is necessary for demonstrating the smart trigger functionality.
- RFID reader: This is the component that will be used to authenticate the user and ensure that only authorized users can fire the gun.
- RFID tag: This is the component that will be attached to the user's ID and will be read by the RFID reader to authenticate the user.
- Microcontroller: This is the brain of the smart trigger system and will be responsible for processing the RFID data and controlling the trigger mechanism.
- Custom PCB: This will be used to connect all the components together and ensure that they work together seamlessly.

Items Excluded:

- Upper receiver: This is the part of the gun that contains the barrel and is not necessary for demonstrating the smart trigger functionality.
- Stock: This is the part of the gun that provides support and is not necessary for demonstrating the smart trigger functionality.
- Grip: This is the part of the gun that is held by the user and is not necessary for demonstrating the smart trigger functionality.

VII. PROJECT TASKS

Describe each task.

- "baseline deliverables" – things you will deliver
- "stretch goals" – things that you are planning to include, but the project will succeed even without them (and why)
- What components will be purchased and what will be built?
- What resources are needed to complete the tasks? These may be your own, or in a lab, or may be people who will help you.

- Task Interfaces: What are the important inputs and outputs of each individual component? Define them.
- Testing: How will each task be tested?
- Design approach: For each component, how will it be designed? What tools will be used (software, hardware, measurement, etc.)
- Current status: What tasks have been finished? What is left to be done?
- Schedule: A date when each task will be completed.
- Parts: Complete parts list with vendor name and information

- Pre-Planning:
- Functional RFID reader:
- software for programming outside of ide
- Successful blocking of fire control
- part fitting
- assembly
- endurance testing

VIII. INTEGRATION STRATEGY

How will the individual tasks/components be integrated? How will the integration be tested?

IX. RISK ASSESSMENT

What are the largest risks to successful demonstration of the project? Describe each risk as green, yellow, or red. At proposal stage, your goal is to have all risks in the "green" category, that is, you have de-risked any major risks to the project. Describe the de-risking process.

X. CONCLUSION

ACKNOWLEDGMENT